

Advanced Microeconomics

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Problem Set 6: Mathematical solutions

1 Application

1) A consumer's utility function is $u = x_1^{1/2} x_2^{1/2}$. She has an income I and prices are p_1 and p_2 . Explore the relationship between utility maximization and expenditure minimization while filling out the values for the given utility function in the table.

Utility Maximizing	Expenditure Minimizing
Marshallian demands: $x_1 = w/2p_1$	Hicksian demands: $h_1 = u(p_2/p_1)^{1/2}$

Plug in the respective demand functions to get the

indirect utility function: $v = w/(4p_1p_2)^{1/2}$	expenditure function: $e = u(4p_1p_2)^{1/2}$
Substitute the expenditure function (e) into the Marshallian demand function to derive the Hicksian demand function: $x_1 = (u(4p_1p_2)^{1/2})/2p_1 = u(p_2/p_1)^{1/2}$	Substitute the indirect utility function (v) into the Hicksian demand function to derive the Marshallian demand function: $h_1 = ((p_2/p_1)^{1/2})(w/(4p_1p_2)^{1/2}) = w/2p_1$
Invert v and replace I by u to get the expenditure function: $v^{-1} = u(4p_1p_2)^{1/2}$	Invert e and replace u by v to get the indirect utility function: $e^{-1} = w(4p_1p_2)^{-1/2}$
Check Roy's identity: $x_1 = -\frac{\frac{\partial v}{\partial p_1}}{\frac{\partial v}{\partial I}} = \frac{2I(p_1p_2)^{1/2}}{4(p_1^3p_2)^{1/2}} = \frac{I}{2p_1}$	Check Shephard's lemma: $h_1 = \frac{\partial e}{\partial p_1} = \frac{u4p_2}{2(4p_1p_2)^{1/2}} = u(p_2/p_1)^{1/2}$

Establish the Slutsky equation:

$$\frac{\partial x_1}{\partial p_1} = \frac{\partial h_1}{\partial p_1} + \frac{\partial x_1}{\partial w} x_1$$
$$-\frac{w}{2p_1^2} = -\frac{w}{4p_1^2} - \frac{w}{4p_1^2}$$